

Quick Check

Lesson 4: Differences Between Acids and Bases Using Commercial Indicators

10 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.1: Acids and Bases | Lesson 4

Before the experiment, Wanjiku predicts what clean tap water will do to litmus paper. Which prediction fits a neutral liquid?

- A** Blue litmus turns red and red litmus stays red
- B** Neither red litmus nor blue litmus changes colour
- C** Both red litmus and blue litmus turn purple
- D** Red litmus turns blue and blue litmus stays blue

In the practical, a group adds 2–3 drops of freshly squeezed lemon juice to a strip of blue litmus paper. What colour does the strip turn?

A It stays blue

B It turns red

C It turns pink

D It turns colourless

A group adds 2 drops of phenolphthalein to 5 drops of liquid handwashing soap solution. What colour do they see?

A Red

B Yellow

C Pink/magenta

D Colourless

A group adds 2 drops of methyl orange to white vinegar (siki). What colour should they record?

- A** Yellow
- B** Orange-yellow
- C** Red
- D** Colourless

Before testing, Otieno predicted that lemon juice would turn phenolphthalein pink. In the experiment, the lemon juice and phenolphthalein mixture stays colourless. What should Otieno write in his results table?

- A** Colourless, because he must record exactly what he sees
- B** Nothing, because the test must have failed
- C** Pale pink, to be halfway between the prediction and the result
- D** Pink, because that is what he predicted

A student tests lemon juice with red litmus paper and sees no colour change. She says, "The litmus did not work." Which explanation is correct?

- A** She did not add enough drops, so the test cannot work
- B** Acids turn litmus red, and red litmus is already red, so no change is seen
- C** Litmus paper only works on bases
- D** Lemon juice is neutral, so nothing happens

A solution is tested with four indicators. Red litmus: no change. Blue litmus: no change. Phenolphthalein: colourless. Methyl orange: orange-yellow. How should the solution be classified?

- A** Acid, because phenolphthalein is colourless
- B** Base, because methyl orange is yellow-coloured
- C** Neutral, because all the indicators show the neutral results
- D** Base, because the litmus papers did not turn red

A student adds phenolphthalein to an unknown solution and it stays colourless. What can the student correctly conclude?

- A** The solution is an acid
- B** The solution is neutral
- C** The solution is not a base, but it could be an acid or neutral, so another indicator is needed
- D** The solution is a base

After today's experiment, which question from earlier lessons can the class move to the 'SOLVED' zone on the Driving Question Board?

- A** Will an acid harm our stomach or corrode metal?
- B** Which of these familiar liquids are acids and which are bases?
- C** Why do acids make some substances fizz?
- D** How can we measure exactly how acidic a liquid is?

Revising your model using today's evidence, which set of labels is correct?

- A** Lemon juice: neutral; baking soda solution: base; tap water: acid
- B** Lemon juice: acid; baking soda solution: base; tap water: neutral
- C** Lemon juice: base; baking soda solution: acid; tap water: neutral
- D** Lemon juice: acid; baking soda solution: neutral; tap water: base

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).